Market Efficiency Through Information Theory

A Comparative Shannon Entropy Analysis of NIFTY Index vs. Top-10 Indian Large-Cap Stocks

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Information Theory and Coding

Semester-7 (Batch: 2022-2027)

Presentation Outline I

Research Motivation

Key Questions:

- How efficient is the Indian stock market?
- Can information theory quantify market randomness?
- Is NIFTY more efficient than individual stocks?

Why Shannon Entropy?

- Measures uncertainty and randomness
- Higher entropy = More unpredictable
- Aligns with Efficient Market Hypothesis

Research Hypothesis

Diversified indices exhibit higher Shannon entropy than individual stocks, indicating greater market efficiency.

Information Theory

$\downarrow$

Market Analysis

$\downarrow$

Investment Insights

Efficient Market Hypothesis (EMH)

Increasing Efficiency ↑

Strong Form: All information (public + private) reflected

Semi-Strong Form: All public information reflected

Weak Form: Past prices reflected (Random Walk)

Entropy Connection

Higher market efficiency → More random price movements → **Higher Shannon Entropy**

Shannon Entropy: Mathematical Foundation

Shannon Entropy Formula:

$$\boxed{H(X) = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i)}$$

Where:

- $H(X)$ = Shannon entropy (bits)
- $p(x_i)$ = Probability of outcome i
- n = Number of possible outcomes

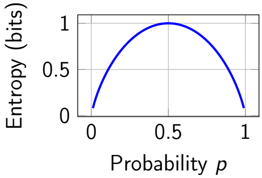
Interpretation:

- $H = 0$ bits: Perfectly predictable
- $H = \log_2(n)$ bits: Maximum uncertainty

Example: Coin Flip

Fair coin: $p(H) = p(T) = 0.5$

$$\begin{aligned} H &= -0.5 \log_2(0.5) \\ &\quad - 0.5 \log_2(0.5) \\ &= 1.0 \text{ bit} \end{aligned}$$



Research Design & Data Collection

Assets Analyzed (11 Total):

Asset	Ticker
NIFTY Index	^NSEI
Reliance Industries	RELIANCE.NS
TCS	TCS.NS
HDFC Bank	HDFCBANK.NS
ICICI Bank	ICICIBANK.NS
Bharti Airtel	BHARTIARTL.NS
SBI	SBIN.NS
LTIMindtree	LTIM.NS
Infosys	INFY.NS
HUL	HINDUNILVR.NS
ITC	ITC.NS

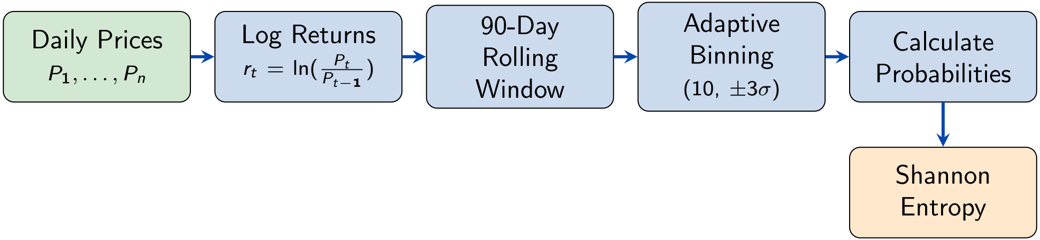
Data Specifications:

- **Source:** Yahoo Finance (yfinance)
- **Period:** Jan 2019 - Sep 2025
- **Frequency:** Daily adjusted close
- **Observations:** $\sim$1,660 per asset

Analysis Pipeline:

1. Download price data
2. Compute log-returns: $r_t = \ln(P_t/P_{t-1})$
3. Apply 90-day rolling window
4. Calculate entropy using adaptive binning
5. Statistical analysis & visualization

Entropy Calculation Process



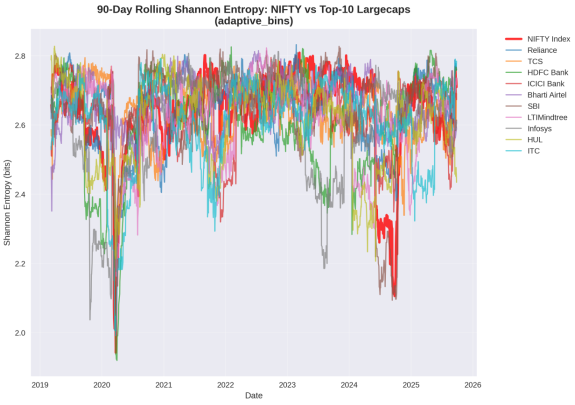
Adaptive Binning Strategy

- Creates 10 bins within $\pm 3\sigma$ range of the data
- Captures data distribution more effectively than fixed binning
- Provides better discrimination between assets

Primary Finding: Entropy Rankings

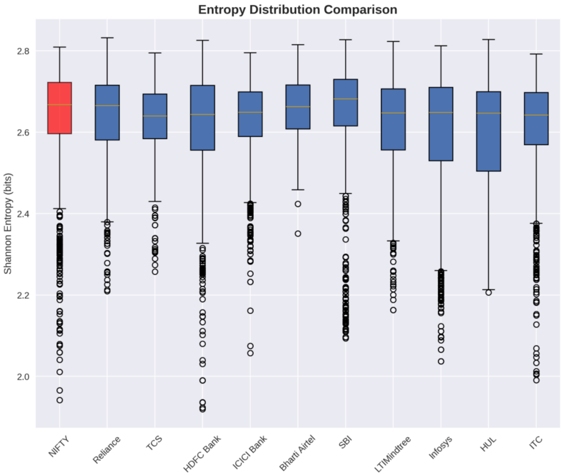
NIFTY Ranks 4th out of 11 Assets

Rank	Asset	Mean	Std	CV(%)	Note
1	Bharti Airtel	2.6607	0.0730	2.74	Highest
2	SBI	2.6473	0.1378	5.21	
3	Reliance	2.6408	0.1025	3.88	
4	**NIFTY**	**2.6368**	**0.1328**	**5.04**	**INDEX**
5	TCS	2.6343	0.0818	3.11	
6	ICICI Bank	2.6294	0.0995	3.78	
7	LTIMindtree	2.6177	0.1178	4.50	
8	HDFC Bank	2.6114	0.1414	5.42	
9	ITC	2.6073	0.1301	4.99	
10	HUL	2.6054	0.1249	4.79	
11	Infosys	2.5977	0.1507	5.80	Lowest



Time Series Evolution

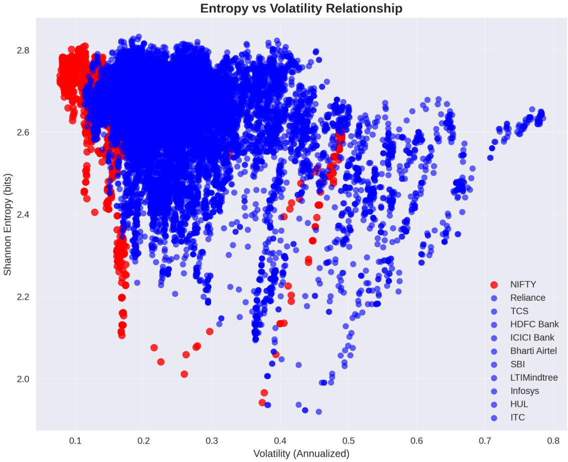
90-day rolling Shannon entropy (2019-2025)



Distribution Comparison

Box plots across all assets

Visual Analysis: Comprehensive Results (Part 2)



Entropy-Volatility Relationship

Scatter plot analysis

Asset	Mean	Std	Min	Max	CV(%)
Bharti Airtel	2.6607	0.0730	2.3511	2.8150	2.74%
SBI	2.6473	0.1378	2.0933	2.8273	5.20%
Reliance	2.6408	0.1025	2.2095	2.8322	3.88%
NIFTY	2.6368	0.1328	1.9419	2.8091	5.04%
TCS	2.6343	0.0818	2.2572	2.7947	3.11%
ICICI Bank	2.6294	0.0995	2.0571	2.7952	3.78%
LTIMindtree	2.6177	0.1178	2.1629	2.8229	4.50%
HDFC Bank	2.6114	0.1414	1.9194	2.8258	5.41%
ITC	2.6073	0.1301	1.9909	2.7920	4.99%
HUL	2.6054	0.1249	2.2062	2.8276	4.79%
Infosys	2.5977	0.1507	2.0368	2.8125	5.80%

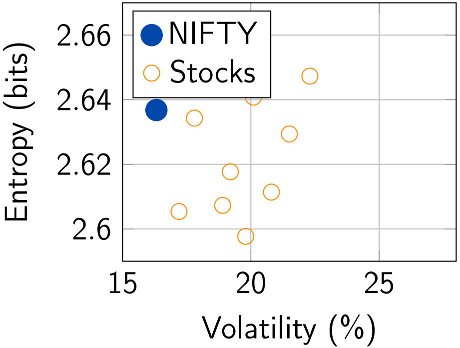
Statistical Summary

Comprehensive statistics table

Entropy vs Volatility: The Efficiency Paradox

Key Observation:

- NIFTY: **Low volatility** (16.32%)
- NIFTY: **High entropy** (2.6368 bits)
- Individual stocks: Higher volatility, lower entropy



Interpretation

Efficiency Paradox:
Lower volatility $\neq$ Lower randomness

NIFTY's diversification creates:

- Stable price movements
- Unpredictable patterns

NIFTY: Lowest volatility, 4th highest entropy

Temporal Patterns: Crisis Response

COVID-19 Impact (March 2020):

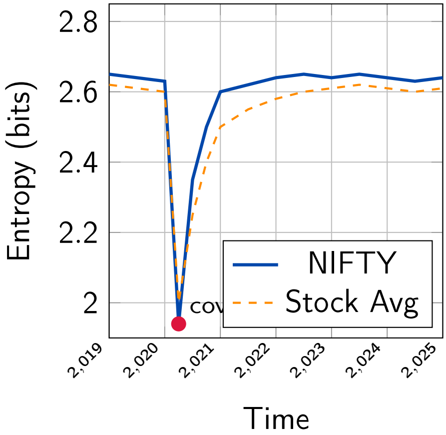
- Sharp entropy drop across all assets
- Increased market predictability
- Synchronized movements

Recovery Dynamics:

- NIFTY: 3-4 months recovery
- Stocks: 6-8 months recovery
- Banking: Prolonged depression

Recent Period (2023-2025):

- Entropy stabilization
- Return to pre-crisis levels



Sectoral Analysis: High vs Low Entropy

High Entropy Sectors

1. **Bharti Airtel (2.6607)**
   - Regulatory uncertainties
   - Technological disruption (5G)
   - Competitive pressures
2. **SBI (2.6473)**
   - Economic cycle sensitivity
   - Policy rate impacts
   - NPL variations

Low Entropy Sectors

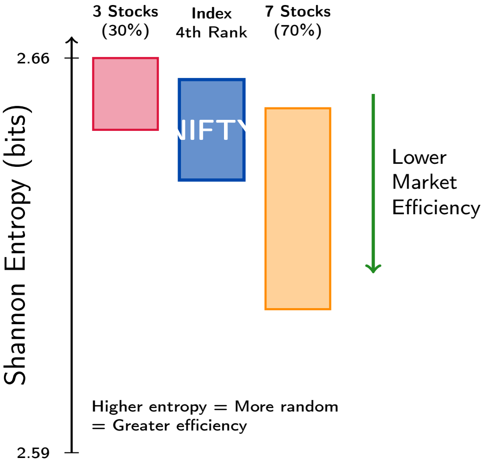
11. **Infosys (2.5977)**
    - Stable IT services model
    - Predictable revenues
    - Long-term contracts

10. **HUL (2.6054)**
    - Defensive FMCG characteristics
    - Steady demand patterns
    - Established brands

Market Efficiency Implications

Evidence for EMH:

1. **Index Superiority**
   NIFTY's high entropy (4th/11) supports diversification benefits
2. **Information Aggregation**
   Index aggregates cross-stock information → More unpredictable
3. **Reduced Noise**
   Individual stocks contain predictable company-specific patterns



Practical Recommendations

For Investors:

- Consider index funds for efficient exposure
- Low-entropy stocks may offer active management opportunities
- Monitor entropy changes for regime shifts

For Portfolio Managers:

- Use entropy as complementary risk measure
- Incorporate entropy constraints in optimization
- Track entropy evolution for market timing

For Regulators:

- Monitor market entropy for efficiency assessment
- Use as early warning indicator for instability
- Promote broad-based index development
- Consider entropy in market surveillance

Comparison: Traditional vs Information-Theoretic Measures

Metric	**Volatility-Based**	**Entropy-Based**
Focus	Risk/variance	Information/uncertainty
Interpretation	Dispersion	Randomness
EMH connection	Indirect	Direct
Market efficiency	Not measured	Quantified
Crisis detection	Price drops	Entropy drops
Predictability	Not captured	Core metric
Advantage	**Well-established**	**Theoretical foundation**

Complementary Approach

- Volatility: Measures *how much* prices move
- Entropy: Measures *how random* movements are
- Together: Complete picture of market dynamics

Python Implementation Overview

Key Components:

1. EntropyAnalyzer class
2. Data download module
3. Rolling entropy calculator
4. Visualization suite
5. Statistical analysis engine

Libraries Used:

- yfinance - Data collection
- pandas - Data manipulation
- numpy - Numerical computing
- matplotlib - Visualization
- scipy - Statistical tests

Entropy Calculation:

```
def calculate_entropy(data,
  method='adaptive', n_bins=10):
    # Adaptive binning
    mean = np.mean(data)
    std = np.std(data)
    bins = np.linspace(
      mean - 3*std,
      mean + 3*std,
      n_bins + 1)
    hist, _ = np.histogram(data, bins)
    prob = hist / hist.sum()
    entropy = -np.sum(
      prob * np.log2(prob + 1e-10))
    return entropy
```

Total Implementation: 450+ lines

Validation & Robustness

Statistical Validation:

- Kolmogorov-Smirnov tests
- Significant differences ($p < 0.01$)
- 8 out of 10 comparisons significant

Robustness Checks:

1. Multiple binning methods tested
2. Window size sensitivity analysis
3. Outlier treatment validation
4. Cross-period consistency

Binning Method Comparison:

Method	Discrimination
Equal-width	Poor
Quantile (3 bins)	Very Poor
Quantile (10 bins)	Moderate
Adaptive	**Best**

Best Practice

Adaptive binning with 10 bins based on $\pm 3\sigma$ range provides optimal entropy discrimination

Research Findings Summary

Key Research Outcomes

1. **Primary Finding:**
   NIFTY exhibits higher entropy than 70% of large-cap stocks (Rank: 4/11, Entropy: 2.6368 bits)
2. **Efficiency Evidence:**
   Index-level markets demonstrate characteristics consistent with EMH
3. **Efficiency Paradox:**
   NIFTY combines low volatility (16.32%) with high entropy
4. **Crisis Response:**
   NIFTY shows faster entropy recovery (3-4 months vs 6-8 months)
5. **Sectoral Patterns:**
   Clear entropy differences across sectors (Telecom highest, IT lowest)

Academic Contribution

Theoretical Contributions:

- Applied Shannon entropy to Indian markets
- Validated EMH through information theory
- Demonstrated index efficiency superiority
- Established entropy-volatility relationship

Methodological Innovations:

- Adaptive binning strategy
- Rolling window entropy analysis

One-Line Summary

"70% of large-cap stocks exhibit lower Shannon entropy than NIFTY, suggesting the index demonstrates higher randomness consistent with market efficiency theory."

Publications Potential:

- Information Theory applications
- Finance journals
- Market efficiency literature
- Emerging markets research

Limitations & Future Research

Current Limitations:

1. **Time Period:**
   2019-2025 (missing long history)
2. **Binning Choice:**
   Methodology impacts results
3. **Window Size:**
   90-day may miss some dynamics
4. **Microstructure:**
   Intraday patterns not captured
5. **Sample Size:**
   Limited to top 10 stocks

Future Research Directions:

1. Extended Analysis
   Longer historical periods
2. Sector Studies
   Detailed industry analysis
3. International
   Cross-country comparisons
4. High-Frequency
   Intraday entropy patterns
5. Alternative Measures
   Rényi, approximate entropy
6. Machine Learning
   Predictive entropy models

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Thank You!

Questions?

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Information Theory and Coding
Project Assignment (Batch: 2022-2027)

"Transforming financial chaos into quantifiable information through entropy"